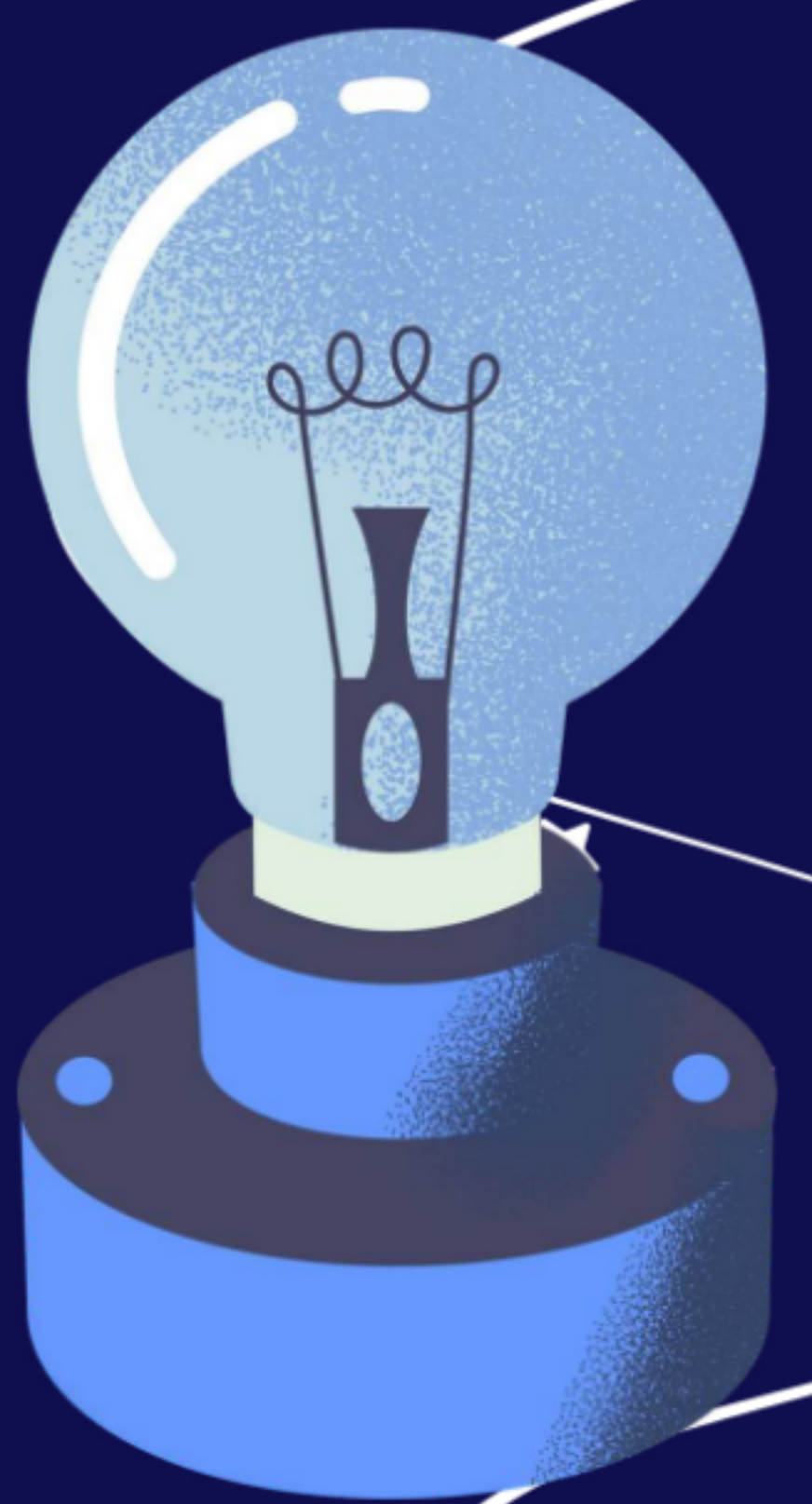


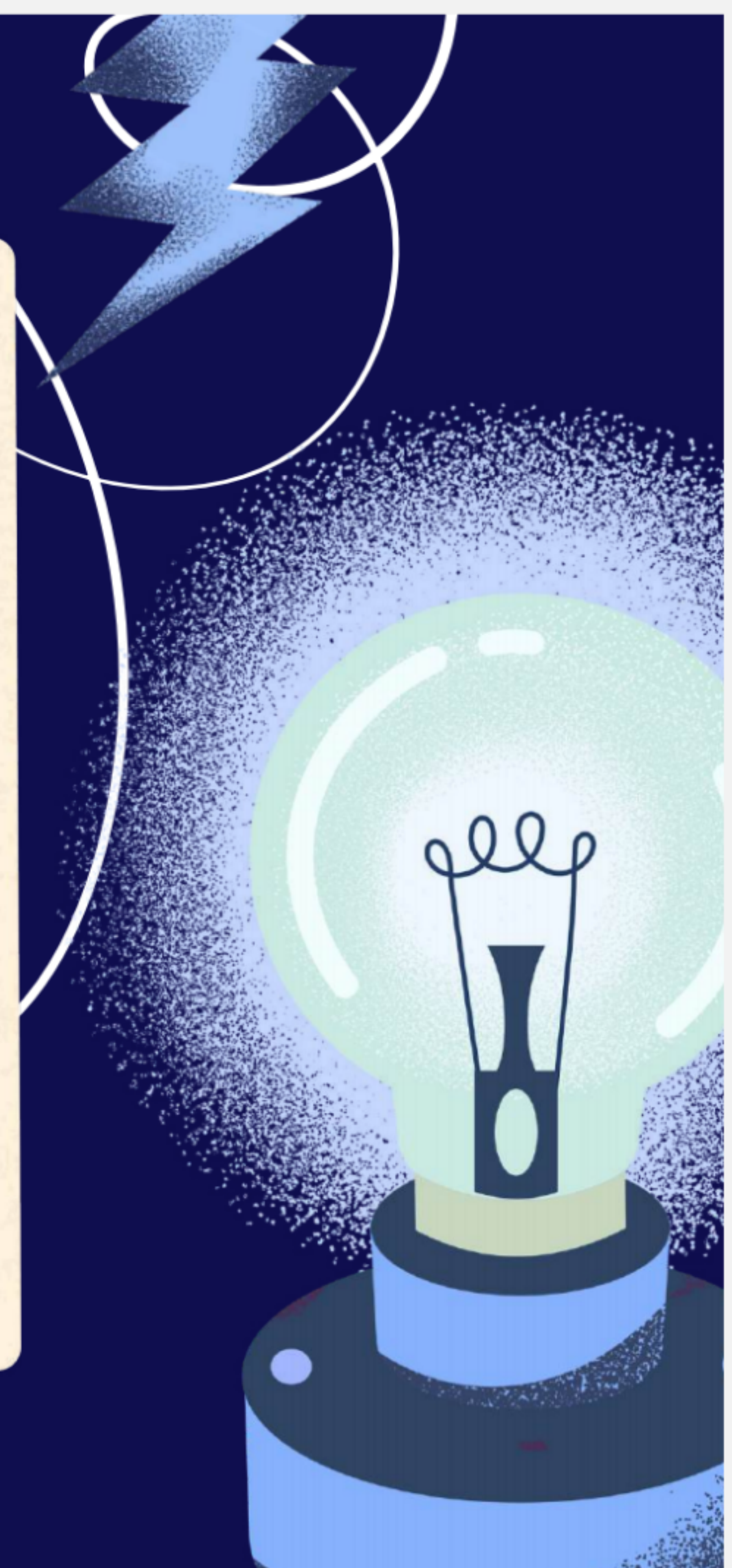
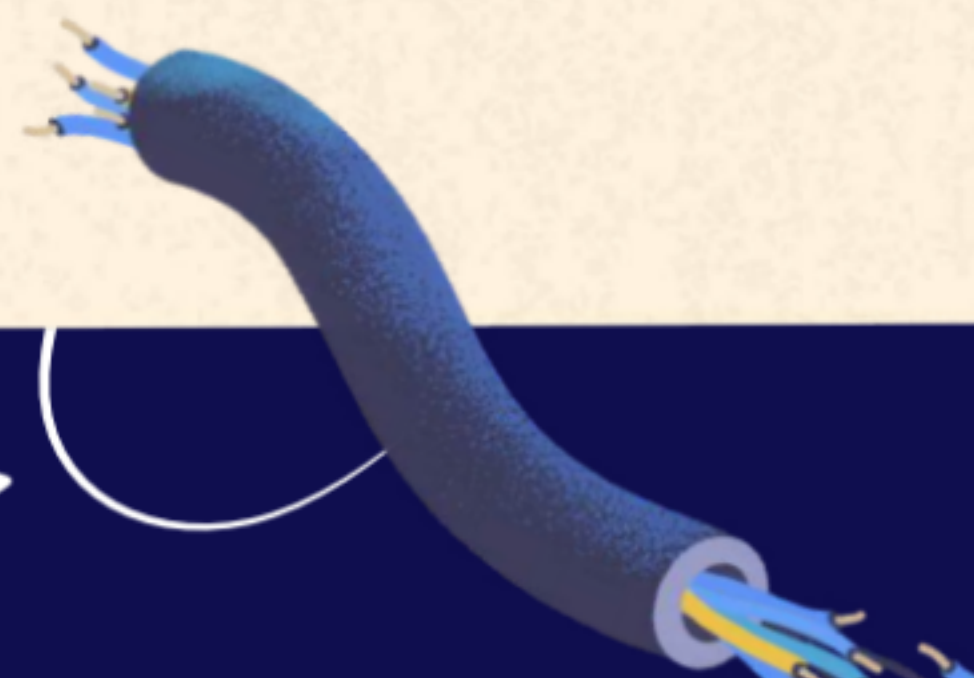


L-3

CLASS X – SCIENCE

ELECTRICITY

PRASHANT KIRAD



JOSH meter

$$V = IR$$
$$P = VI = I^2R = \frac{V^2}{R}$$
$$E = Pt$$



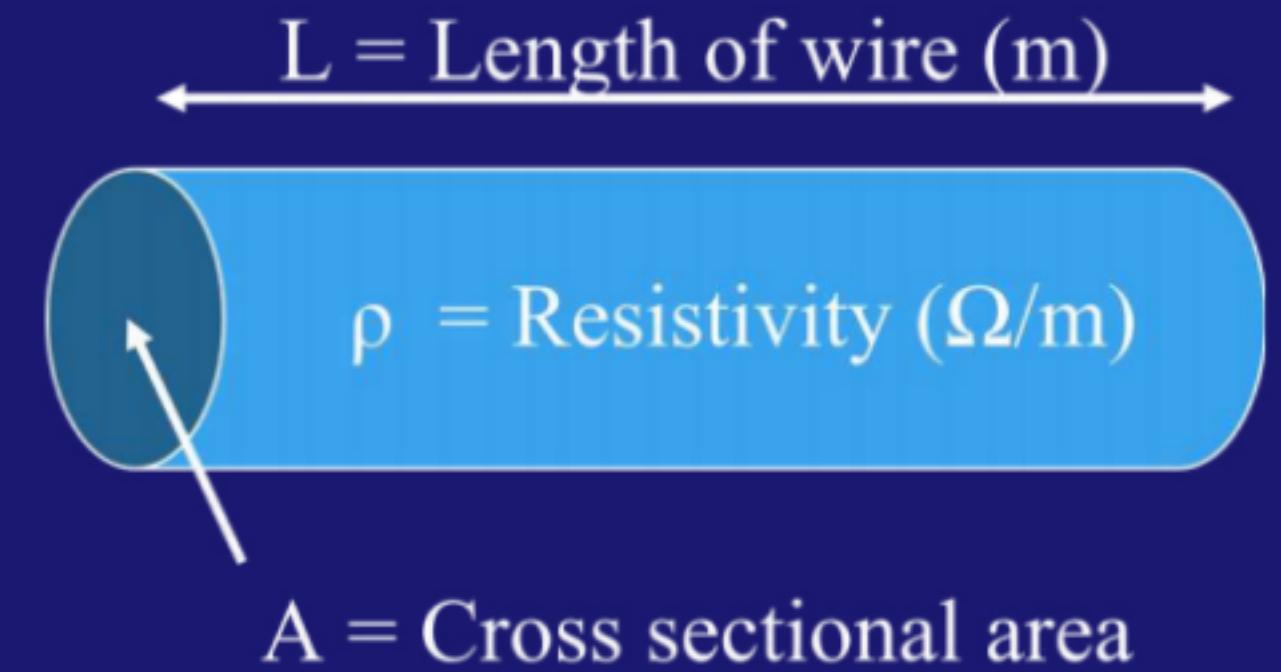
RESISTIVITY

- Resistance of a conductor is directly proportional to its length, inversely proportional to its area of cross-section, and depends on the material of the conductor. **It is denoted by ρ .**

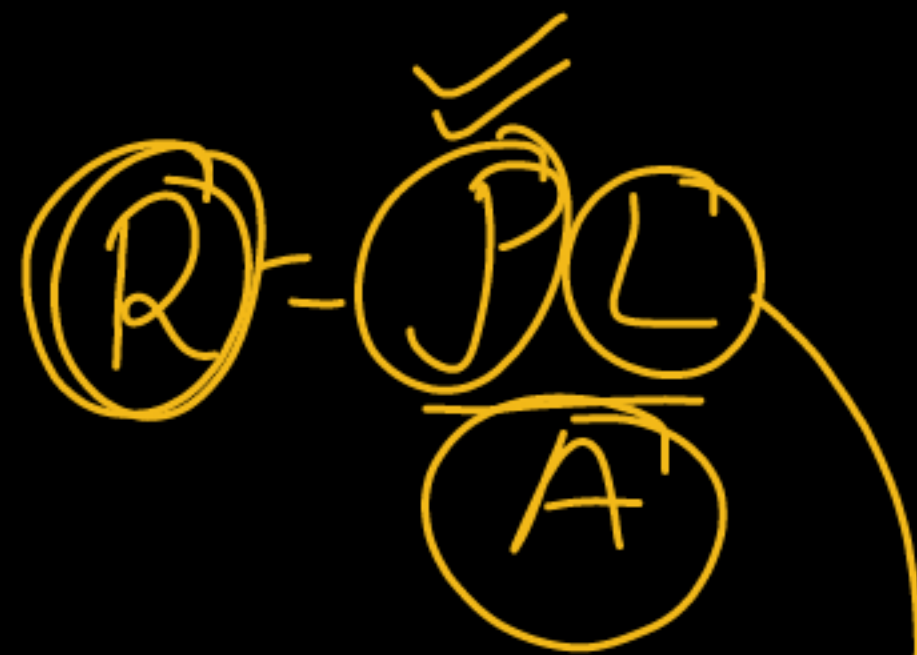
$$R = \rho L / A$$

$$\rho = RA / L$$

$$\rho = \Omega m$$



- Resistivity does not change with change in length or area of cross-section but it changes with change in temperature.***
 - Tungsten $\rightarrow$ filament of bulbs (high melting point).
 - Copper, aluminium $\rightarrow$ transmission wires (low resistivity).



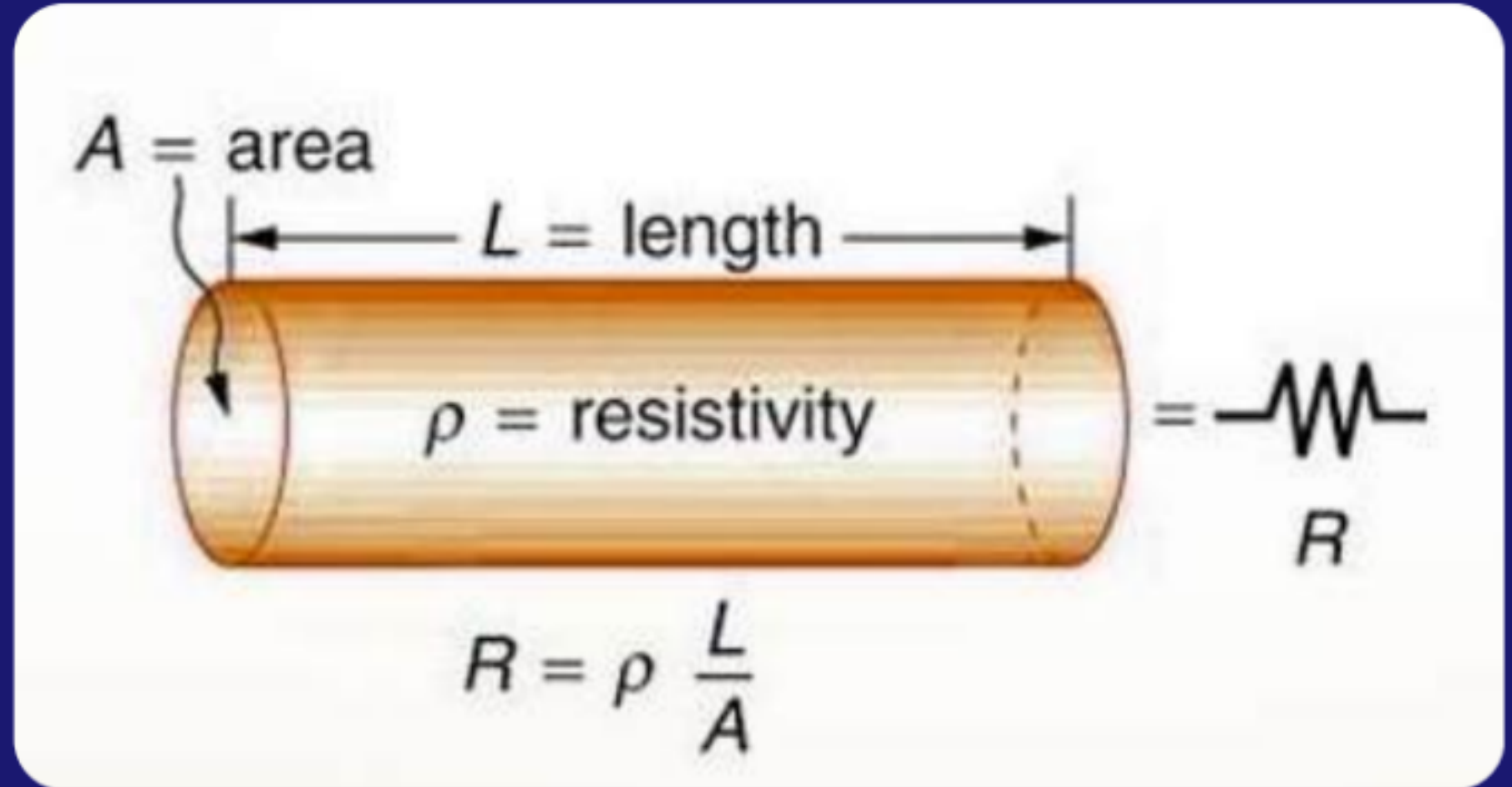
$$\Omega = \frac{P(m)}{m^2}$$

$$\boxed{\Omega_m = P}$$

DIFFERENCE BETWEEN RESISTANCE & RESISTIVITY



V//S



DIFFERENCE BETWEEN RESISTANCE & RESISTIVITY

Resistance (R)	Resistivity (ρ)
Opposition offered by a conductor to the flow of electric current.	The resistance of a conductor of unit length and unit cross-section.
Resistivity is independent of the length and area of cross-section of the conductor, but it depends on the nature of the material and its temperature.	Independent of length and area of cross-section. Depends only on the nature of the material.
Formula: $R = \rho \times (L / A)$	Formula: $\rho = R \times (A / L)$
SI unit: ohm (Ω)	SI unit: ohm-metre ($\Omega \cdot m$)

RESISTIVITY OF ELECTRICAL SUBSTANCES

	Material	Resistivity ($\Omega \text{ m}$)
Conductors	Silver	1.60×10^{-8}
	Copper	1.62×10^{-8}
	Aluminium	2.63×10^{-8}
	Tungsten	5.20×10^{-8}
	Nickel	6.84×10^{-8}
	Iron	10.0×10^{-8}
	Chromium	12.9×10^{-8}
	Mercury	94.0×10^{-8}
	Manganese	1.84×10^{-6}
	Constantan (alloy of Cu and Ni)	49×10^{-6}
Alloys	Manganin (alloy of Cu, Mn and Ni)	44×10^{-6}
	Nichrome (alloy of Ni, Cr, Mn and Fe)	100×10^{-6}
Insulators	Glass	$10^{10} - 10^{14}$
	Hard rubber	$10^{13} - 10^{16}$
	Ebonite	$10^{15} - 10^{17}$
	Diamond	$10^{12} - 10^{13}$
	Paper (dry)	10^{12}

[NCERT]

Q. Resistance of a metal wire of length 1 m is $26\ \Omega$ at 20°C . If the diameter of the wire is 0.3 mm, what will be the resistivity of the metal at that temperature? predict the material of the wire.

$$R = \frac{\rho L}{A}$$

$$\rho = \frac{RA}{L}$$

=

$$\checkmark R = 26\ \Omega$$

$$\checkmark L = 1\text{ m}$$

$$D = 0.3\text{ mm} \rightarrow 0.03\text{ cm} \rightarrow 0.0003\text{ m}$$

$$A = \pi r^2$$

$$= \frac{22}{7} \times \left(\frac{3 \times 10^{-4}}{2} \right)^2$$

Q. In case of four wires of same material, the resistance will be minimum if the diameter and length of the wire respectively are:

(a) $D/2$ and $L/4$

(b) $D/4$ and $4L$

(c) $2D$ and L

(d) $4D$ and $2L$

$D/4$

$R_{min} = ?$

$$R_3 \propto \frac{L}{\pi D^2}$$

$$R_1 \propto \frac{2L}{\pi (2D)^2}$$

$$R_4 \propto \frac{2L}{4\pi D^2} = \frac{L}{2\pi D^2}$$

$$R_1 \propto \frac{L}{\frac{\pi}{4} D^2}$$

$$R_1 \propto \frac{L}{\pi D^2} \times 4$$

$$R_1 \propto \frac{4L}{\pi D^2}$$

$$R \propto \frac{L}{A}$$

$$R_2 \propto \frac{4L}{\pi \left(\frac{D}{8}\right)^2} = \frac{4L \times 64}{\pi D^2} = \frac{256L}{\pi D^2}$$

$$\frac{\frac{2}{3} \times \frac{6}{4}}{\frac{4}{6}}$$

$$\Rightarrow \frac{2 \times 4}{\frac{3}{4}}$$

$$\Rightarrow \frac{\frac{2}{4}}{10} = \frac{2}{5}$$

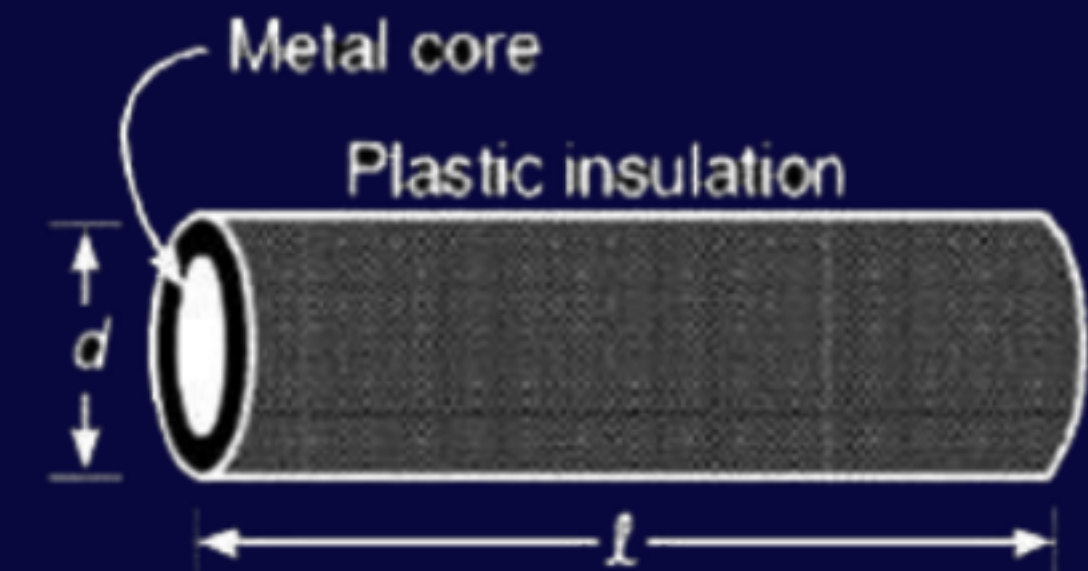
Q. Plastic insulation surrounds a wire having diameter d and length l as shown above. A decrease in the resistance of the wire would be produced by an increase in the:

(A) length l of the wire

(B) diameter d of the wire $\rightarrow A \uparrow R \downarrow$

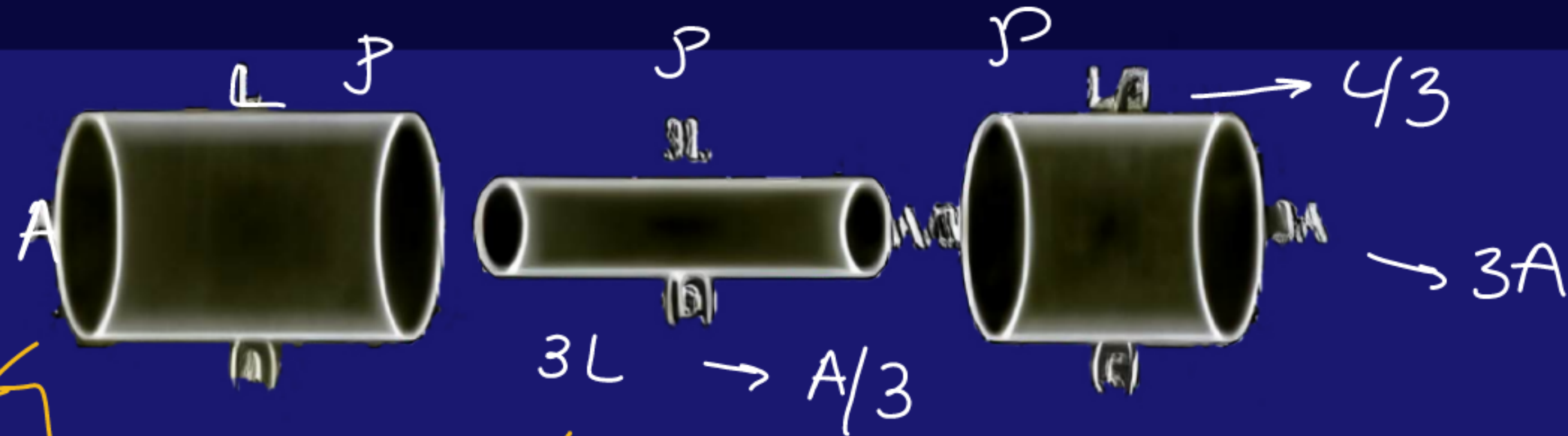
(C) temperature of the wire

(D) thickness of the plastic insulation



$R \downarrow \rightarrow$

Q. The three conductors shown in the figure are made of the same material. Compare their resistances and resistivities. Justify your answer.



$$R_1 = \frac{\rho L}{A}$$

$$\frac{R_1}{R_2} = \frac{1}{9}$$

$$R_2 = \frac{\rho 3L}{A/3} = \frac{9\rho L}{A}$$

$$R_2 : R_3 = 81 : 1$$

$$R_3 = \frac{\rho L/3}{3A} = \frac{\rho L}{9A}$$

[NCERT]

Q. Will current flow more easily through a thick wire or a thin wire of the same material, when connected to the same source? Why?

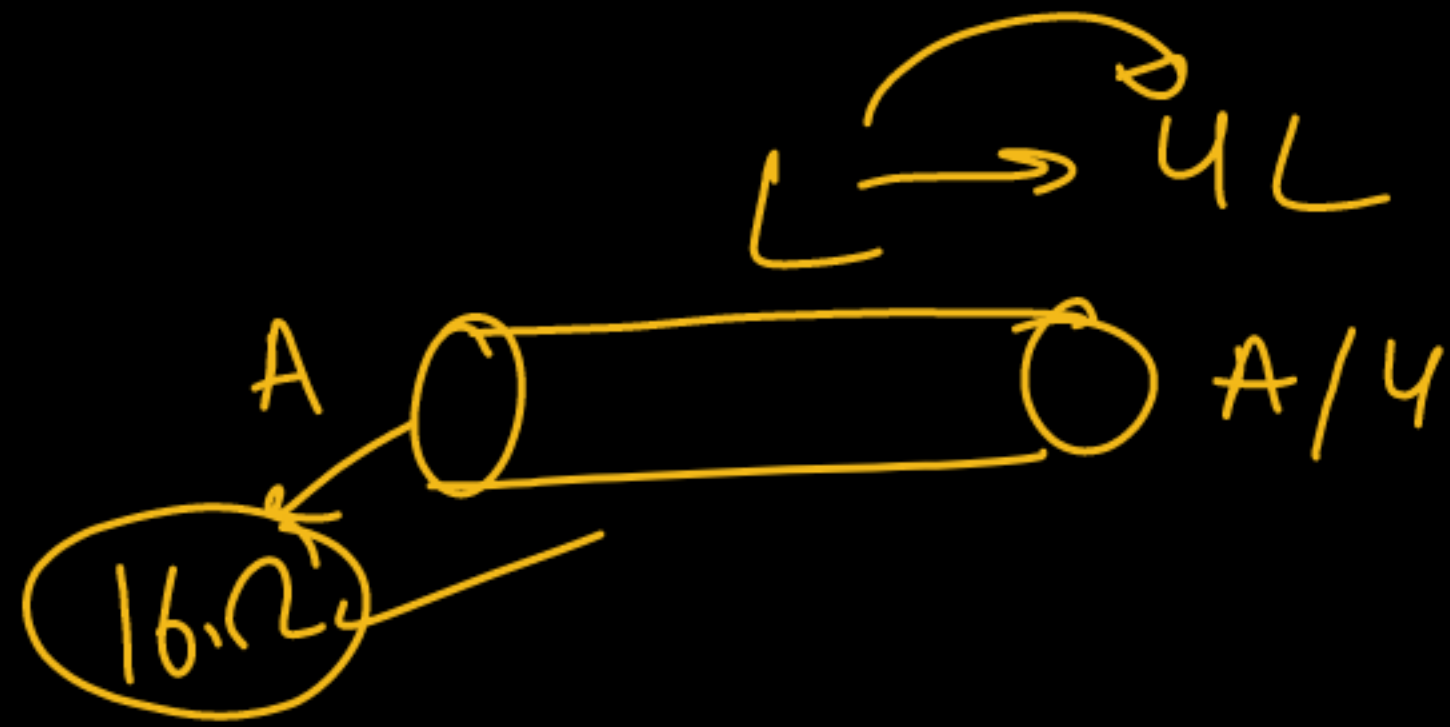
Current will flow more easily through a thick wire of the same material.

Reason: A thick wire has a larger area of cross-section, so it offers less resistance to the flow of current.

Since,

$$R \propto 1/A$$

As area increases, resistance decreases. Therefore, current flows more easily through a thick wire.



$$R' = \frac{P \cdot 4L}{\frac{A}{4}} = 16 \left(\frac{P \cdot L}{A} \right) = 16 \times 16 = 256 \Omega$$

Q. Nichrome wire of length l and radius r has resistance of $10\ \Omega$. How would the resistance of the wire change when:

(i) Only length of the wire is doubled?

(ii) Only diameter of the wire is doubled?

Justify your answer.

$l \rightarrow 2l$ $\rightarrow 20\ \Omega$

$\downarrow R \propto \frac{1}{A} \uparrow$

$R = \frac{10}{4} = 2.5\ \Omega$

$A = \pi \left(\frac{D}{2}\right)^2$

$A \rightarrow 4 \text{ times}$

$R \propto \frac{1}{D^2} \rightarrow 3 \text{ times}$
 $A \rightarrow 9 \text{ times}$

[NCERT]

Q. A wire of given material having length l and area of cross-section A has a resistance of $4\ \Omega$. What would be the resistance of another wire of the same material having length $l/2$ and area of cross-section $2A$?

$$R = 4\ \Omega = \frac{\rho L}{A}$$

$$R' = \frac{\rho \frac{L}{2}}{2A} = \frac{\rho L}{4A} = \frac{1}{4} \left(\frac{\rho L}{A} \right) = \frac{1}{4} \times 4 = 1\ \Omega$$

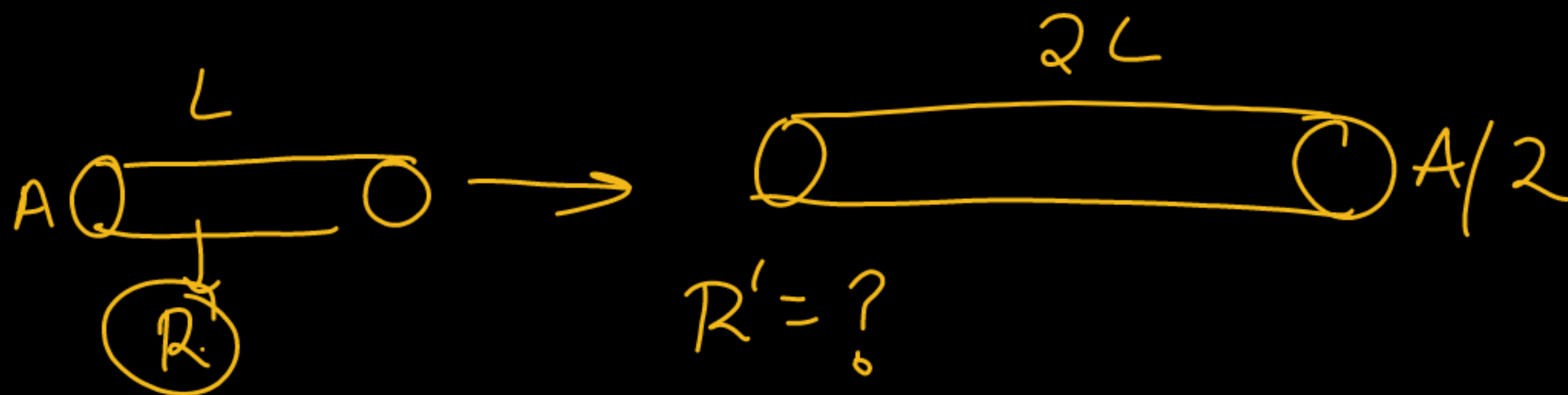
[NCERT]

Q. Why are coils of electric toasters and electric irons made of an alloy rather than a pure metal?

Coils of electric toasters and electric irons are made of an alloy rather than a pure metal because:

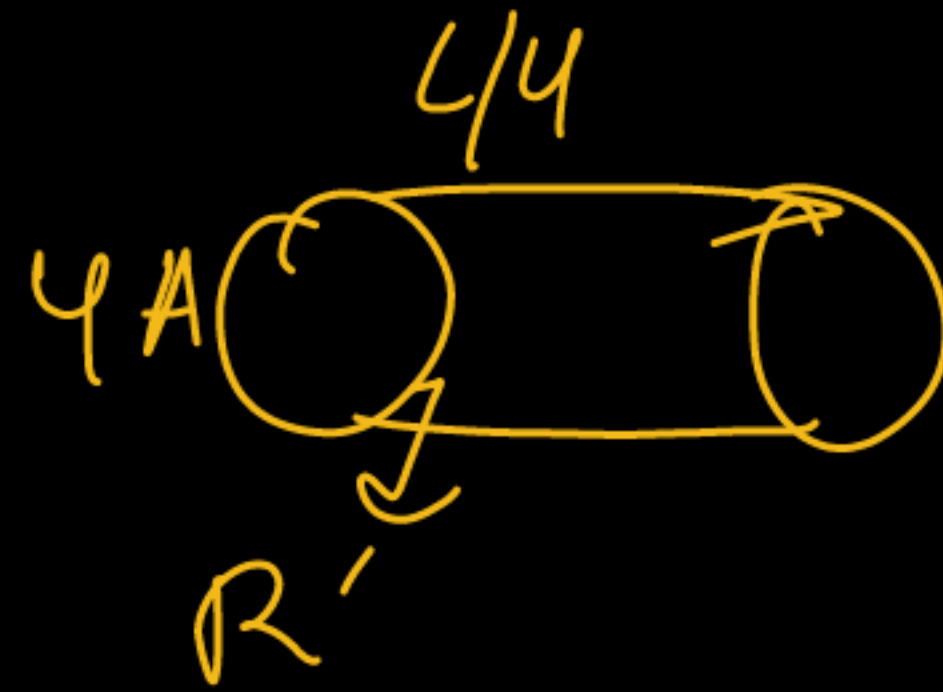
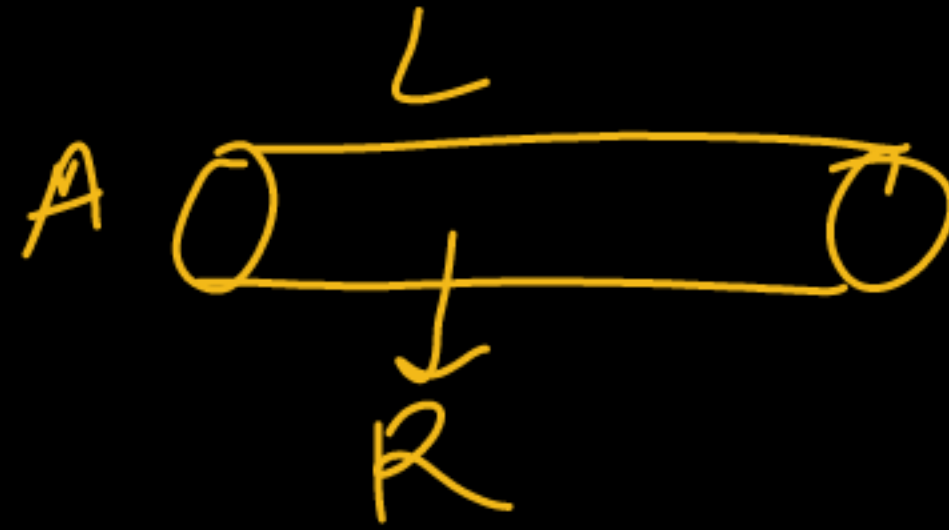
- ✓ 1. Alloys have high resistivity, so they produce more heat.
- ✓ 2. Alloys do not oxidise or burn easily at high temperature.
- ✓ 3. They can withstand high temperature for a long time.

Therefore, alloys like nichrome are used in heating devices such as electric toasters and electric irons.



$$R = \frac{P(2L)}{\frac{A}{2}} = 4 \left(\frac{PL}{A} \right) = 4R$$

#



$$R' = \frac{P L}{4}$$

$$\frac{R'}{4A} = \frac{1}{16} \left(\frac{P L}{A} \right) R$$

$$R' = \frac{R}{16}$$

$$\begin{array}{c} \textcircled{V} = A \times \textcircled{L} \rightarrow 2L \\ \downarrow \\ \frac{A}{2} \end{array}$$

Q. At 20°C , the resistivities of three materials A, B and C are:

$$\mathbf{A = 1.6 \times 10^{-8} \Omega m}$$

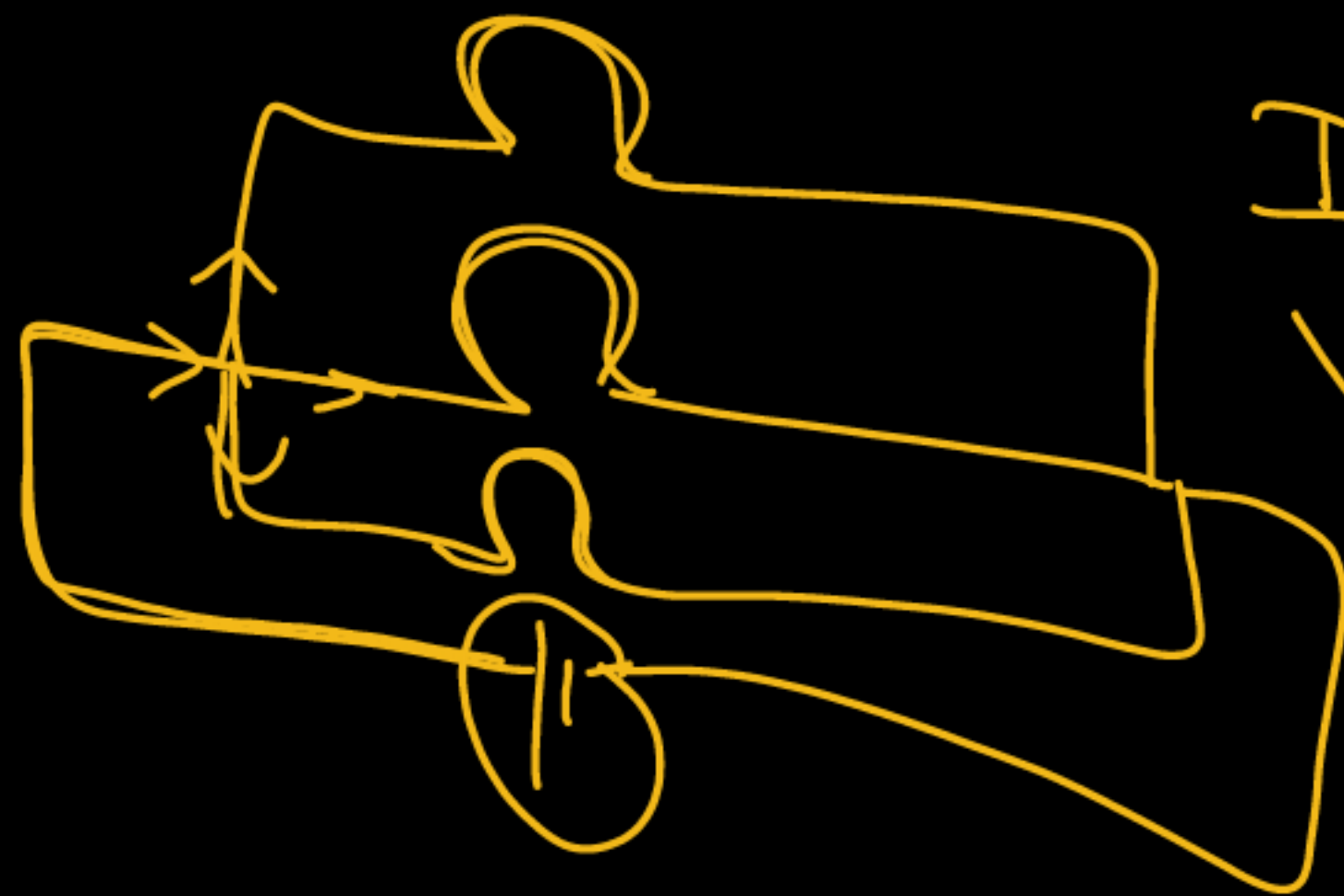
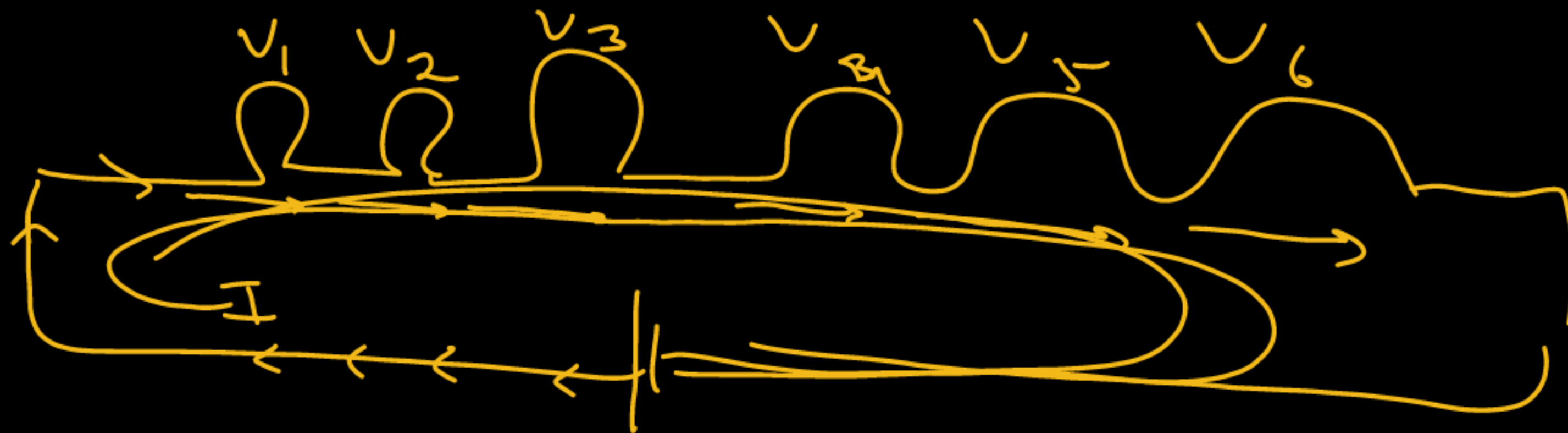
$$\mathbf{B = 100 \times 10^{-6} \Omega m}$$

$$\mathbf{C = 10^{12} \Omega m}$$

Identify them as a metal, an alloy and an insulator. Which material can be used for making heating elements?

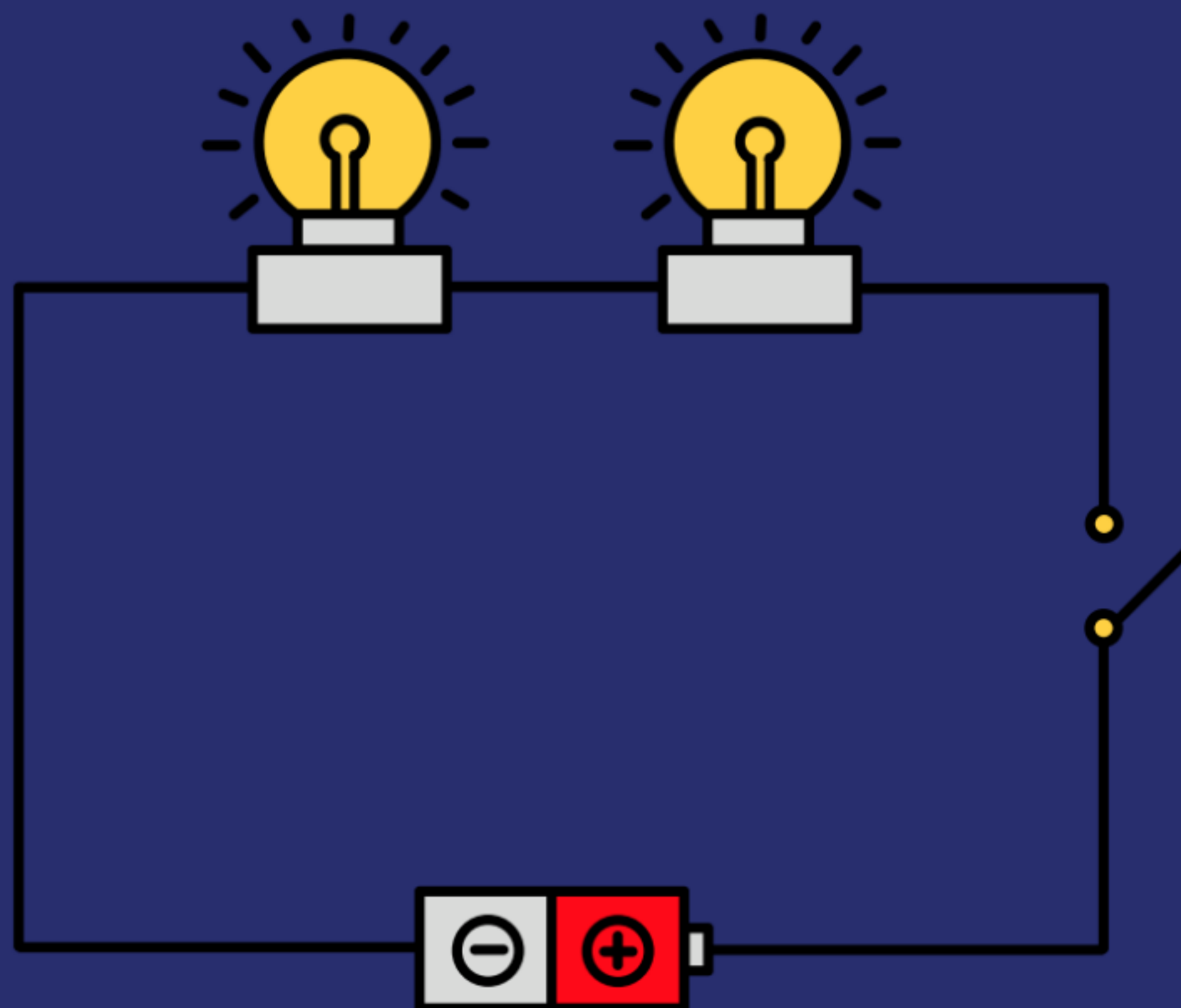
HW

Series $\begin{cases} I \rightarrow \text{same} \\ V \rightarrow \text{different} \end{cases}$



$\begin{cases} I \rightarrow \text{different} \\ V \rightarrow \text{same} \end{cases}$

Series Circuit



SERIES CIRCUIT

In series circuit:

Current remains same throughout all resistors.
Voltage divides itself on all the resistors in proportion to the value of resistance.

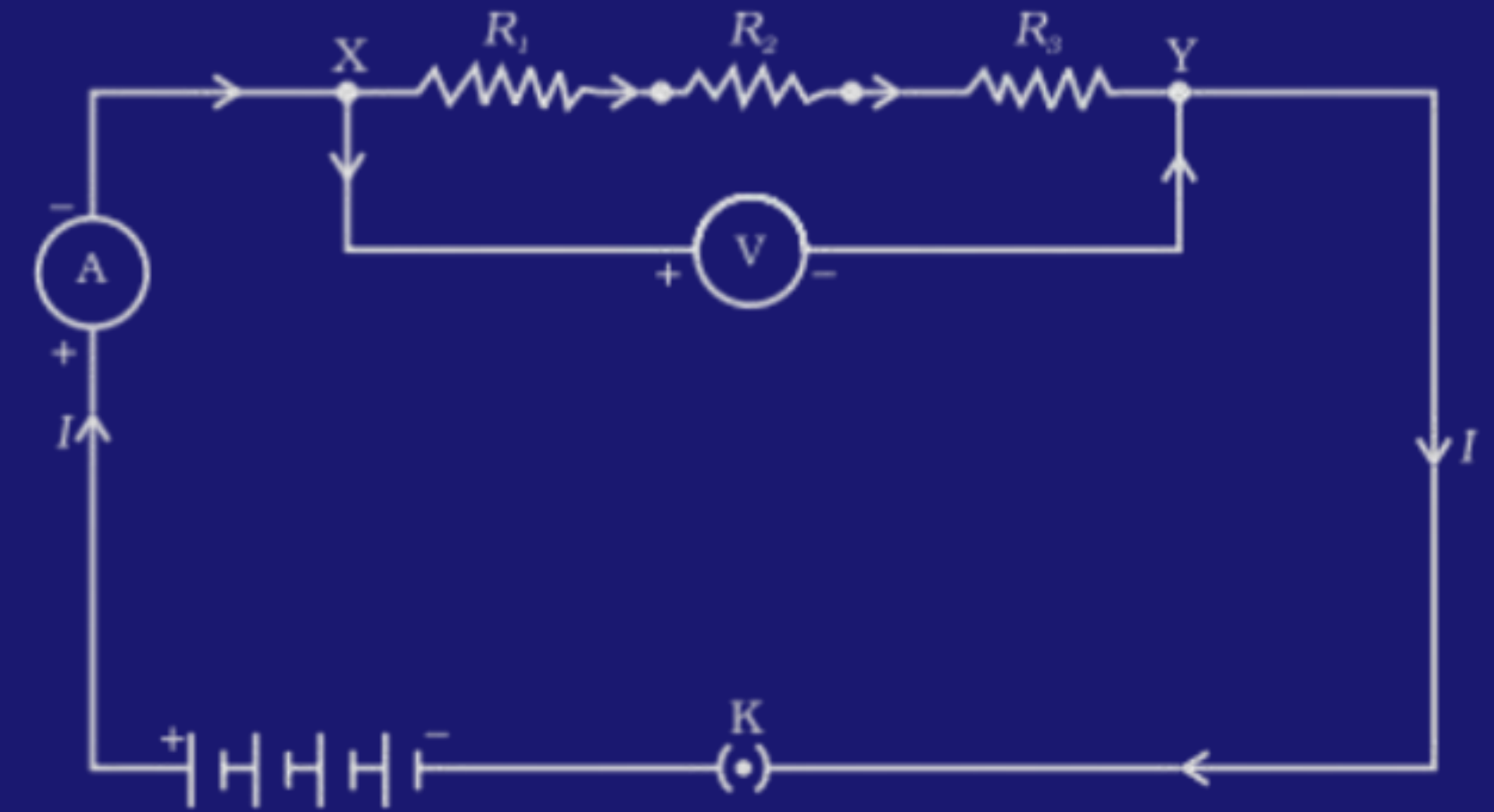


Figure 12.6 Resistors in series

The total potential difference across a series of resistors equals the sum of the potential differences across each individual resistor.

$$V = V_1 + V_2 + V_3$$

Since we know, $V = IR$

$$V_1 = IR_1$$

$$V_2 = IR_2$$

$$V_3 = IR_3$$

$$IR = IR_1 + IR_2 + IR_3$$

$$\Rightarrow R = R_1 + R_2 + R_3$$

$$V = \underbrace{V_1}_{\downarrow} + \underbrace{V_2}_{\downarrow} + \underbrace{V_3}_{\downarrow}$$

$$\downarrow [V = IR]$$

$$IR_{eq} = IR_1 + IR_2 + IR_3$$

$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = \text{net Resistance}$$

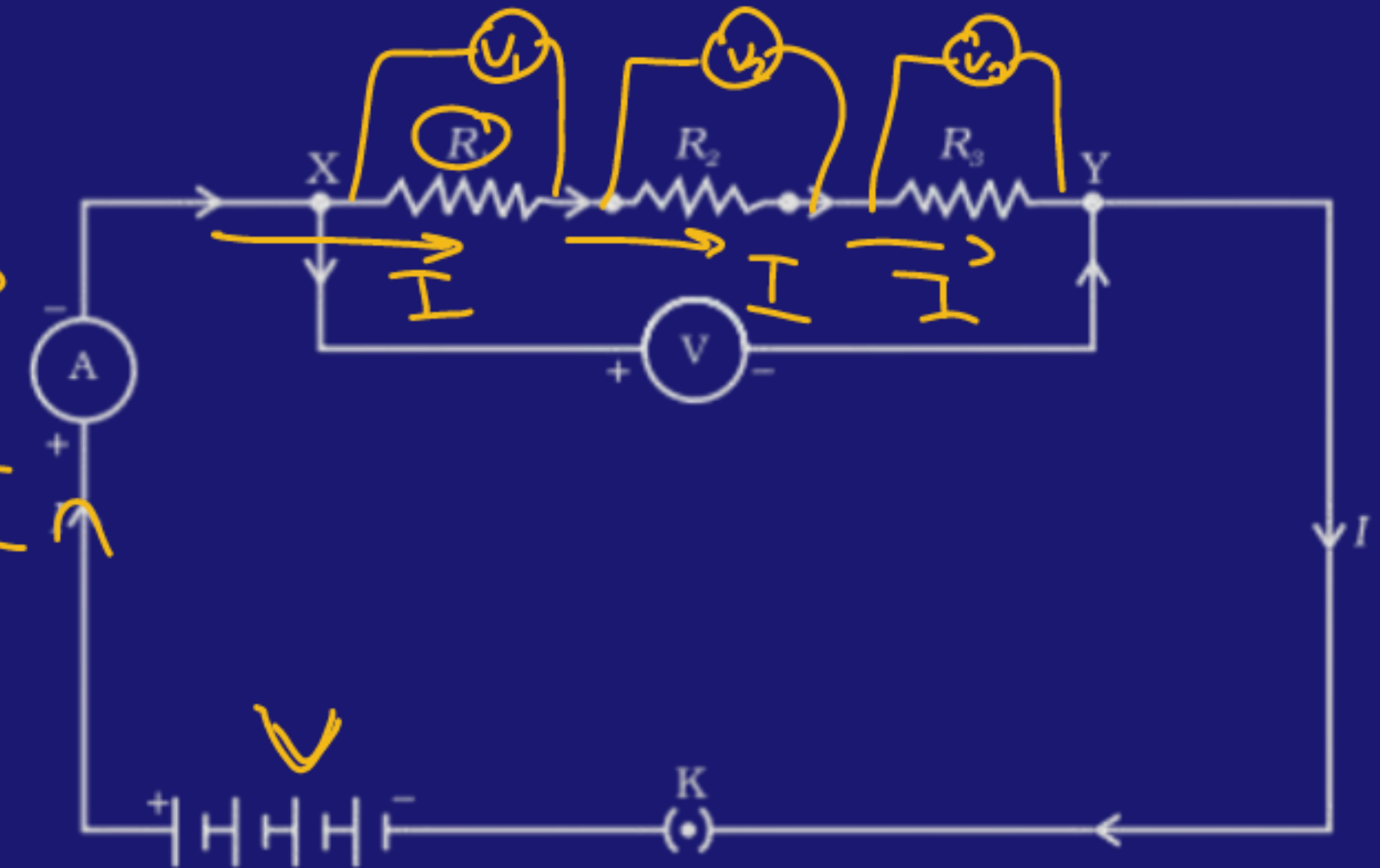
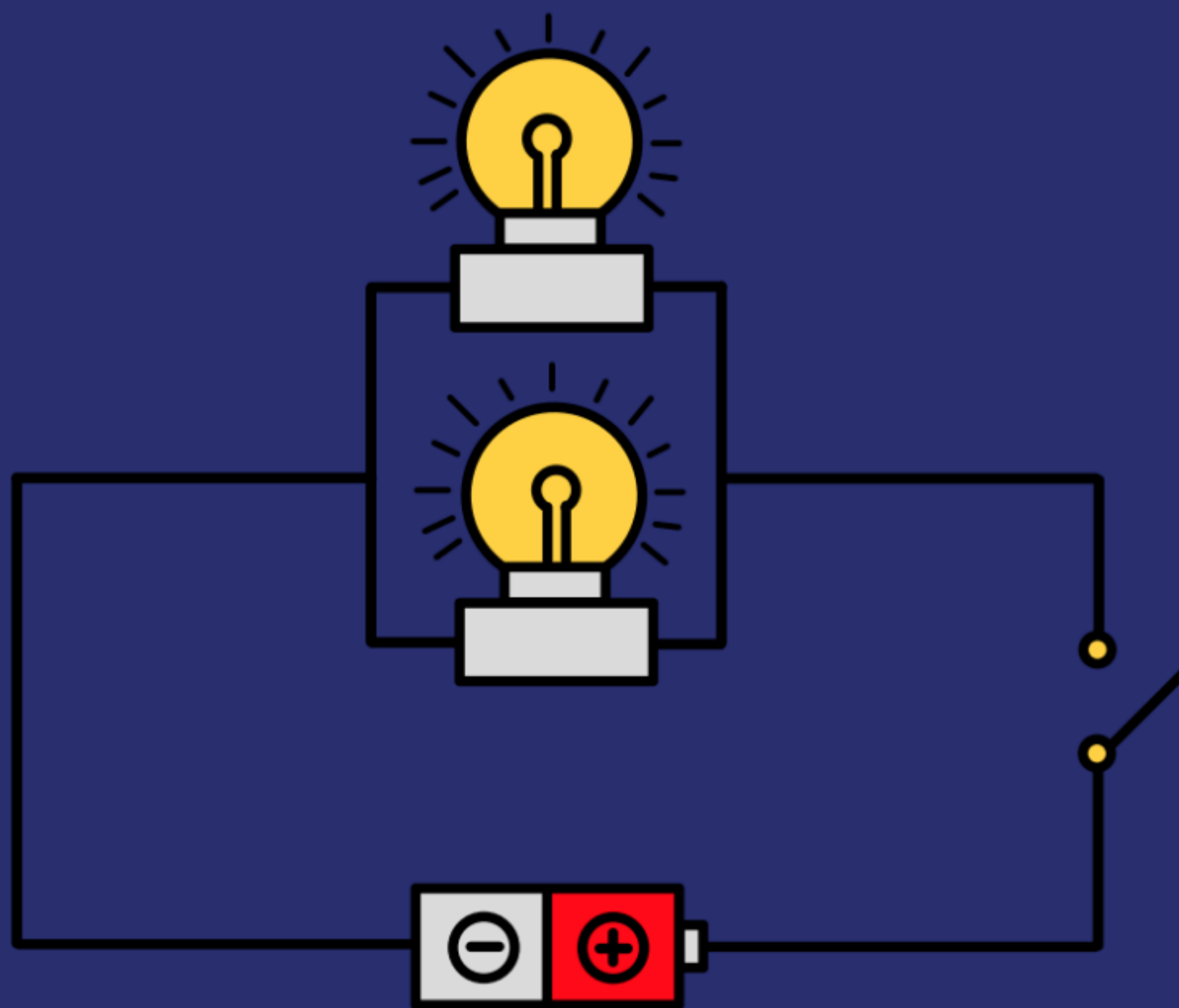


Figure 12.6 Resistors in series

Parallel Circuit



PARALLEL CIRCUIT

In Parallel Circuit :

- Current is divided inversely proportional to the resistance.
- Voltage remains the same.

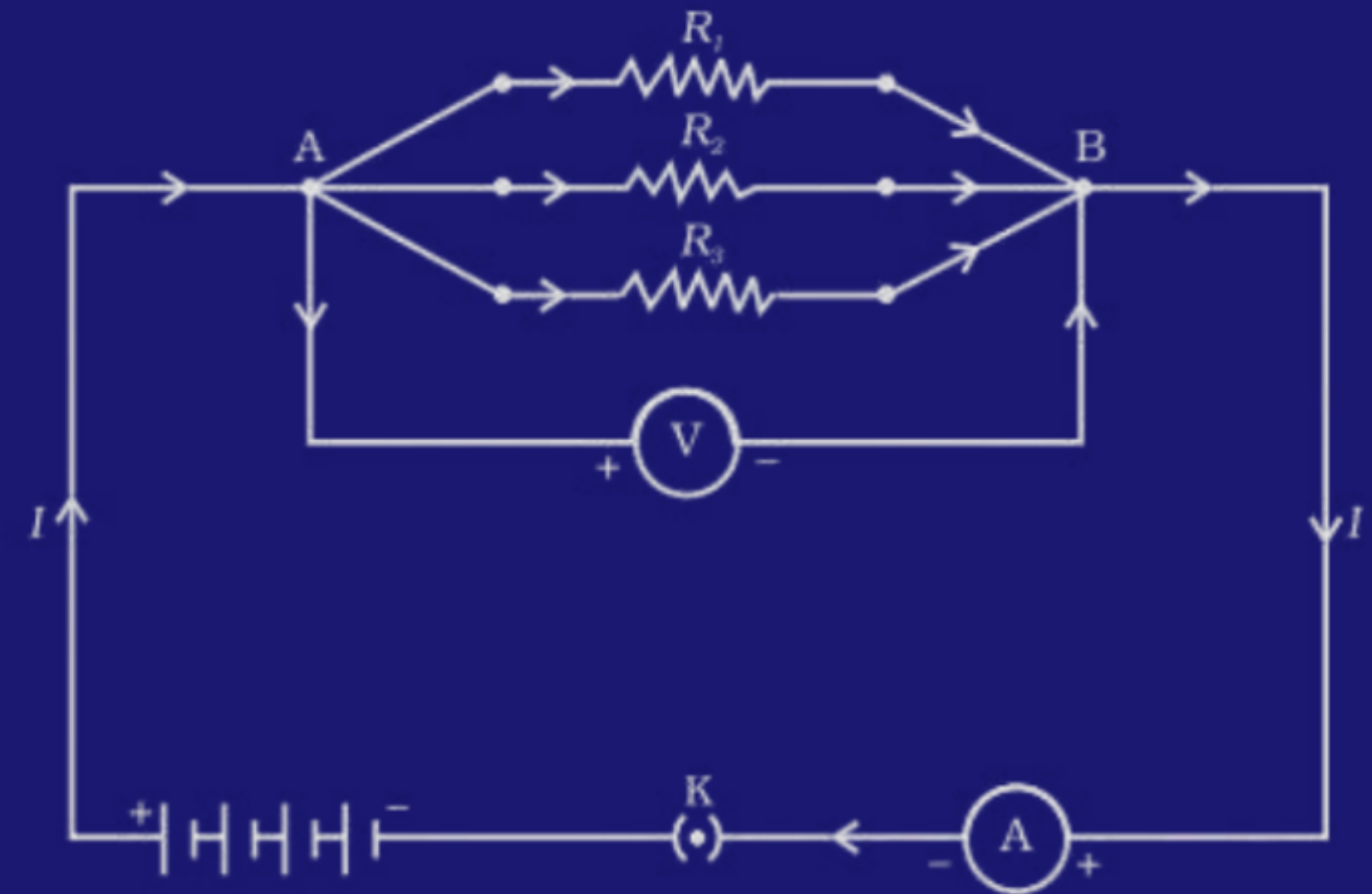


Figure 12.7 Resistors in parallel

The total current I , is equal to the sum of the separate currents through each branch of the combination..

$$I = I_1 + I_2 + I_3$$

Since we know, $I = V / R$

$$I_1 = V / R_1$$

$$I_2 = V / R_2$$

$$I_3 = V / R_3$$

Substitute the values:

$$V / R_p = V / R_1 + V / R_2 + V / R_3$$

Cancel V from both sides:

$$1 / R_p = 1 / R_1 + 1 / R_2 + 1 / R_3$$

$$I = I_1 + I_2 + I_3$$

$$\left[\begin{array}{l} V = IR \\ I = \frac{V}{R} \end{array} \right]$$

$$\frac{V}{R_p} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

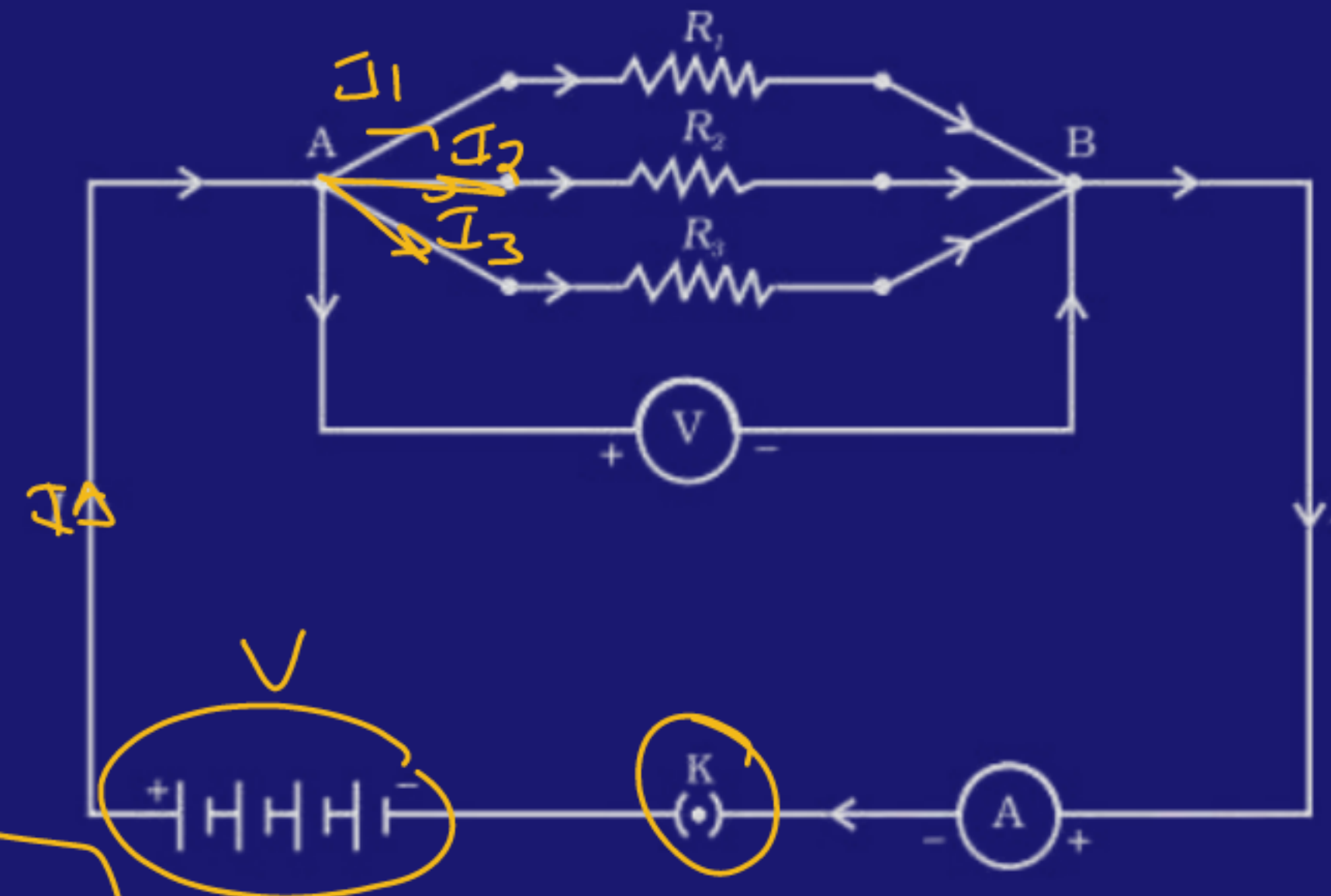
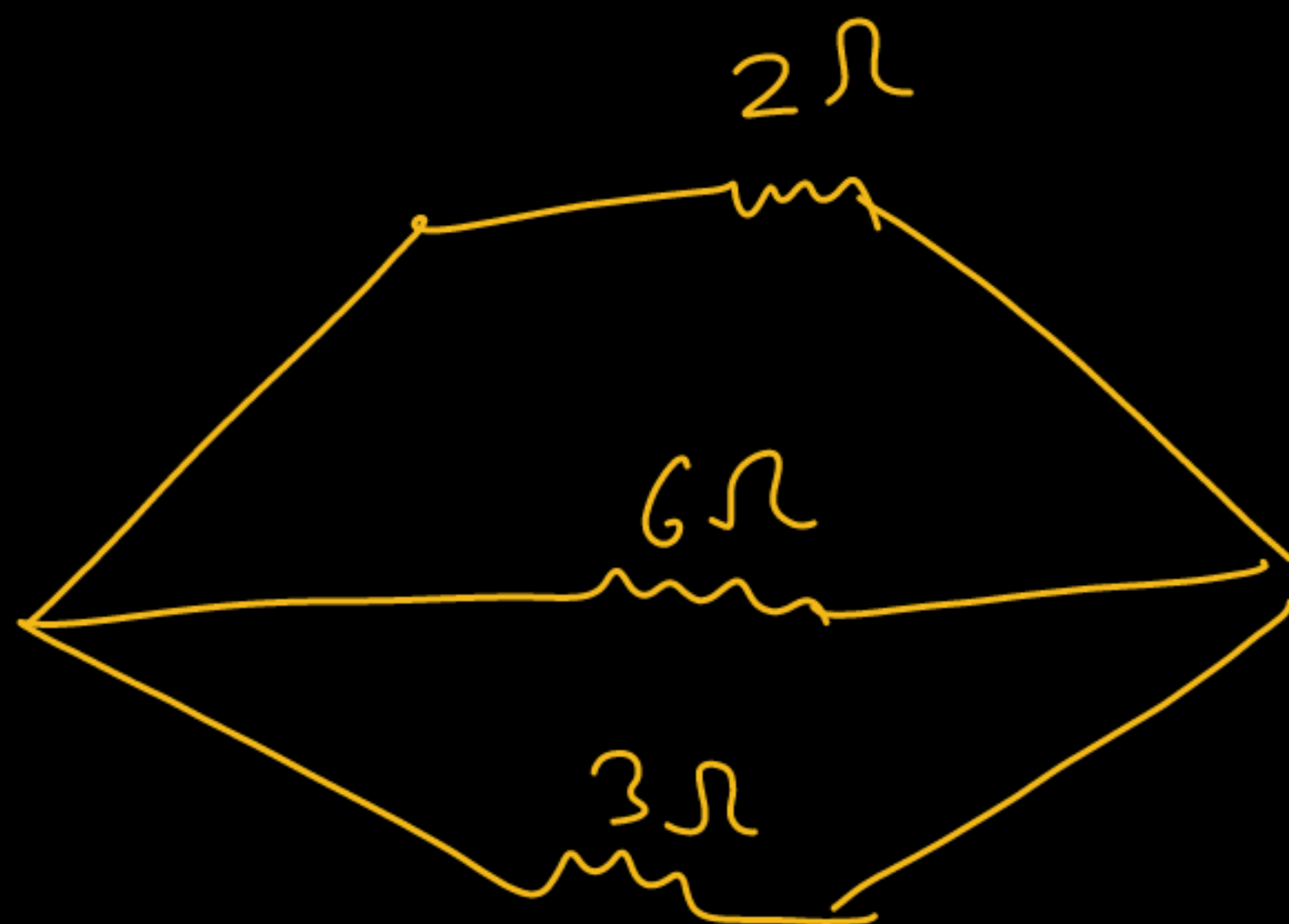


Figure 12.7 Resistors in parallel

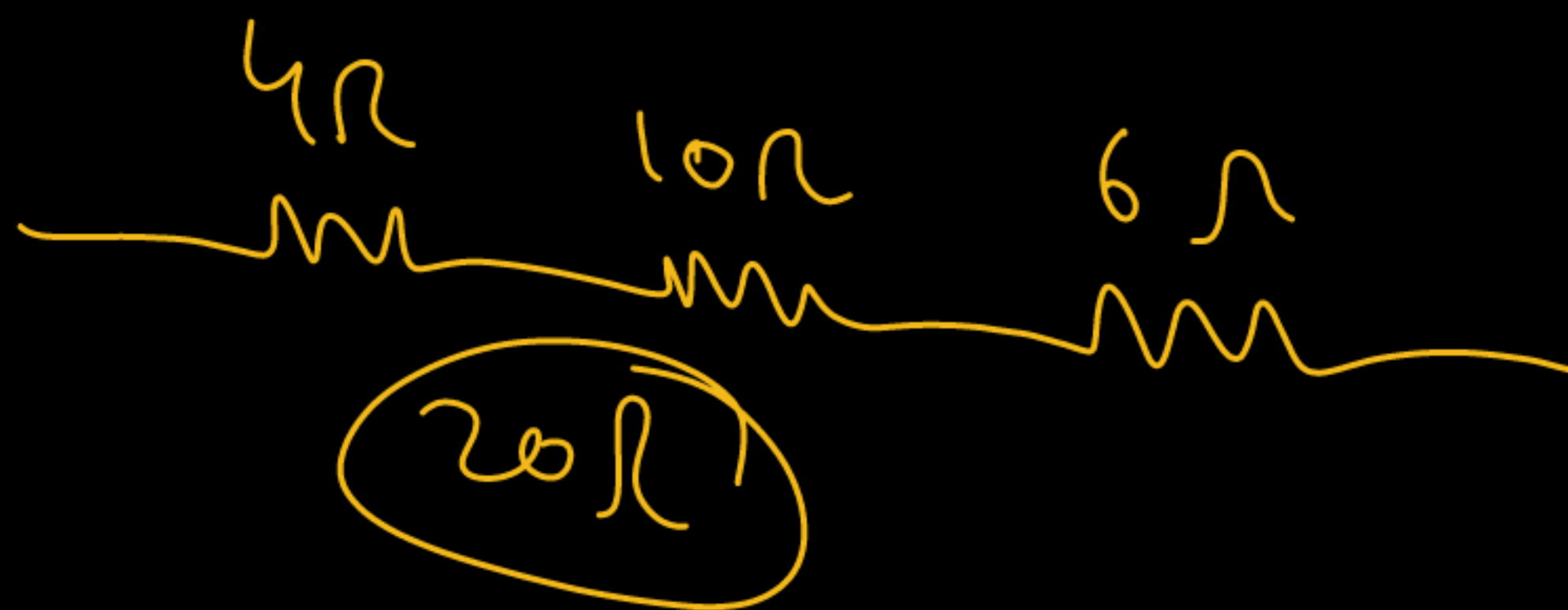


$$\frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3}$$

$$\frac{1}{R_{eq}} =$$

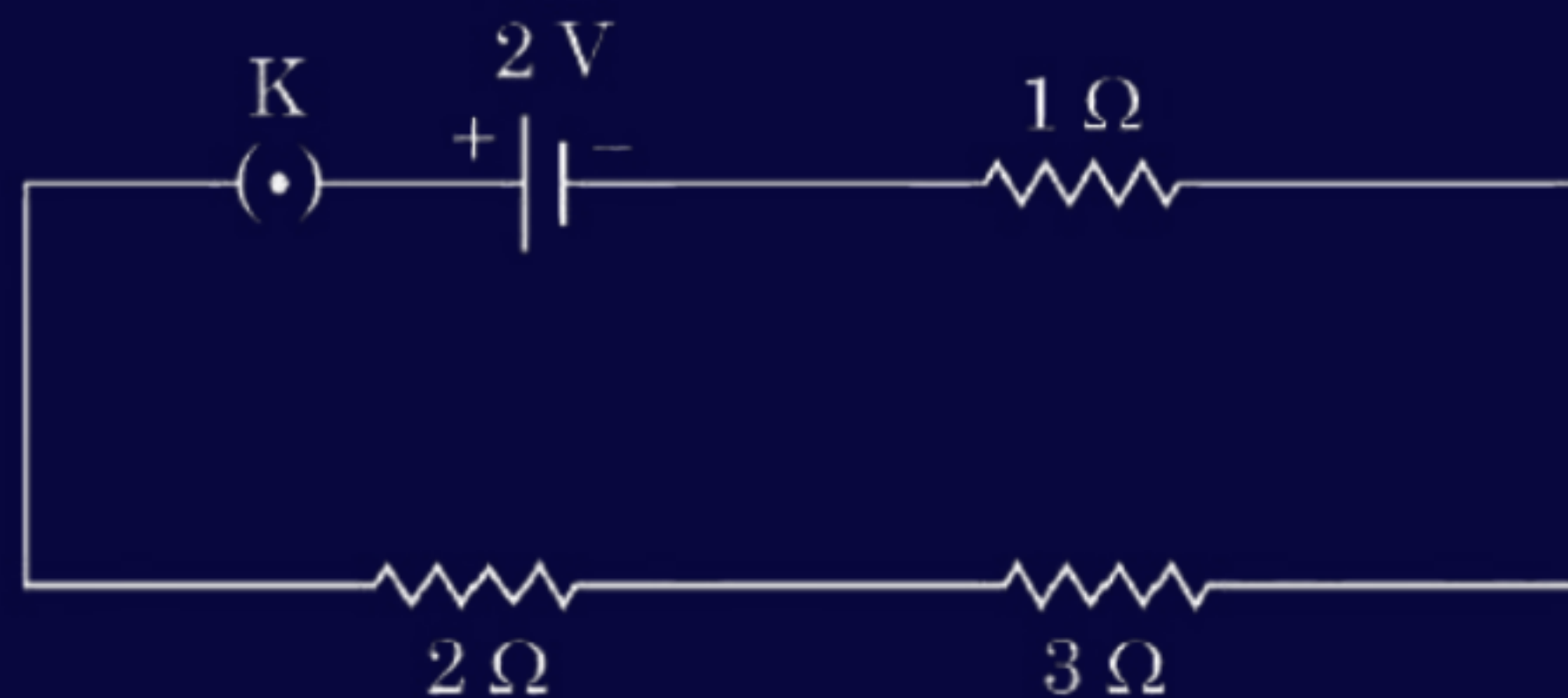
$$\frac{3+1+2}{6} = \frac{6}{6} = 1$$

$$1$$



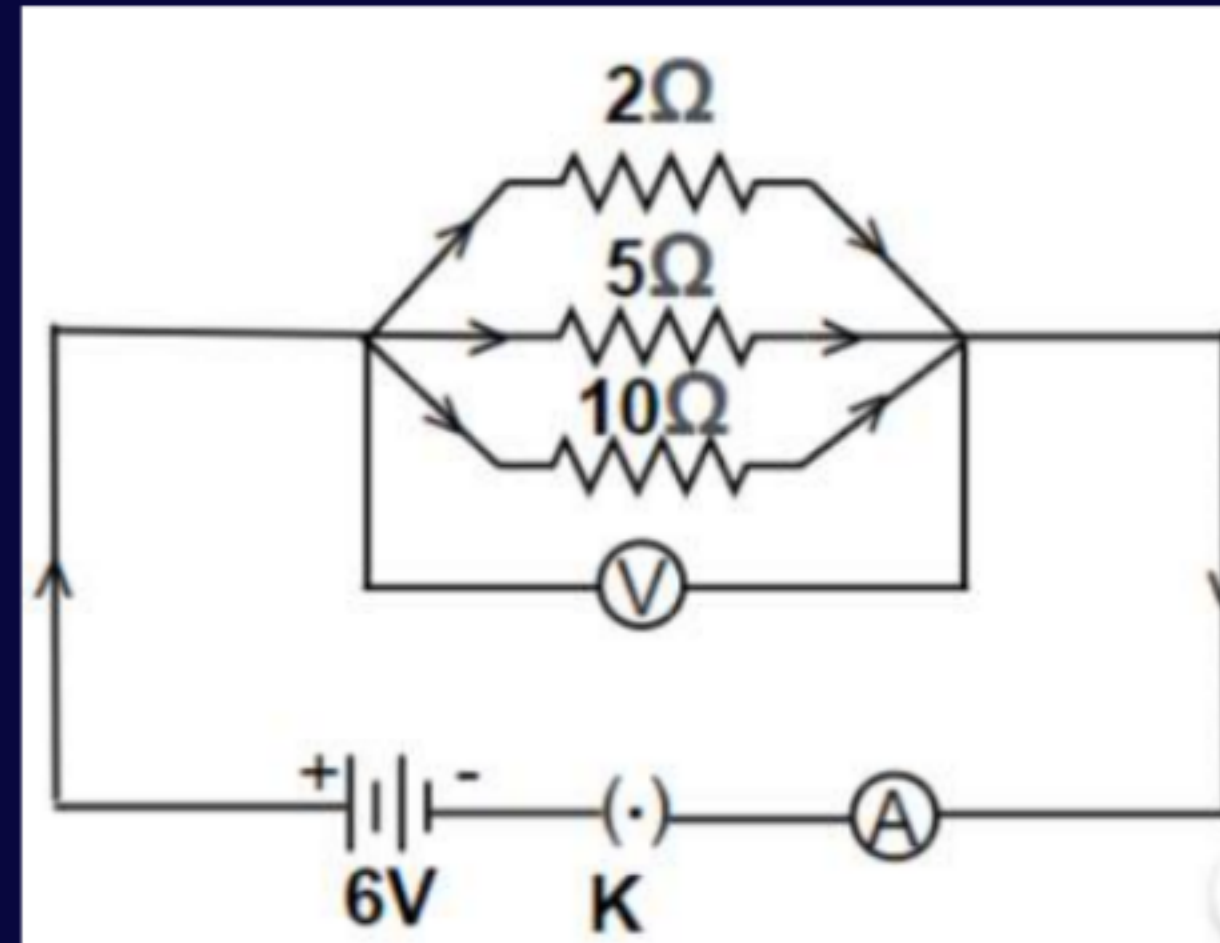
Q.3

The circuit shown below consists of a 2 V battery, a key K, and three resistors of resistances $1\ \Omega$, $2\ \Omega$, and $3\ \Omega$ connected in series. Calculate the total effective resistance of the circuit.



Q. Calculate the total effective resistance of the circuit shown in the diagram.

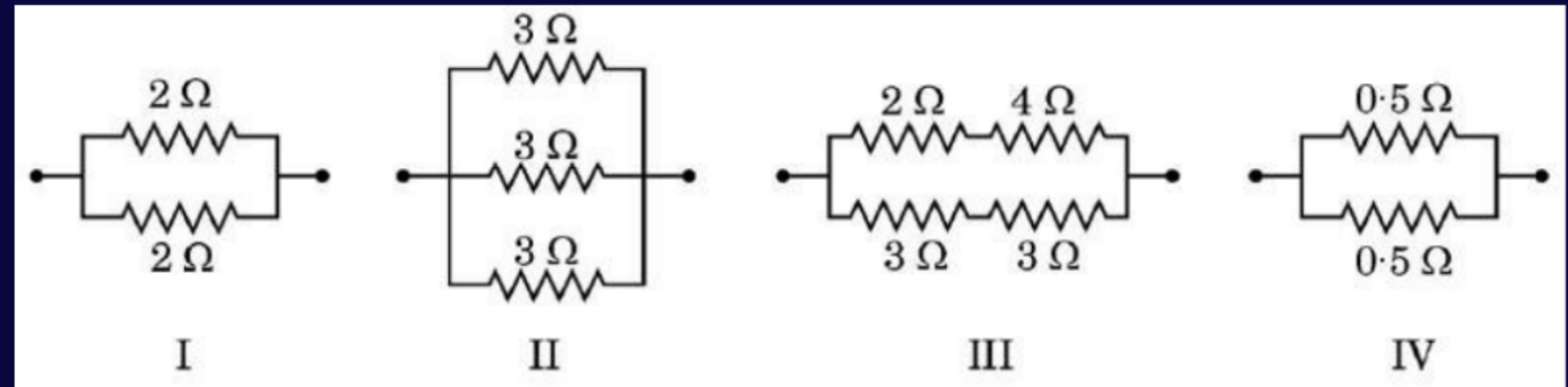
HW



Q. Consider the following combinations of resistors: (1 Mark)

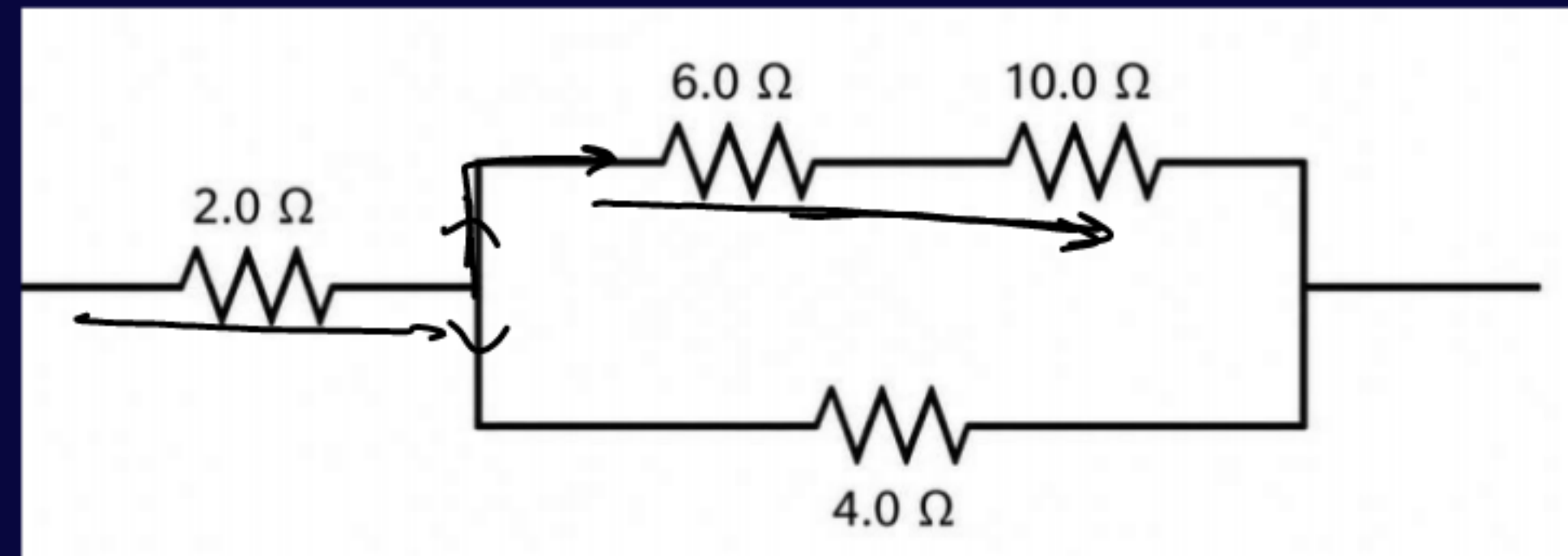
The combinations having equivalent resistance $1\ \Omega$ is/are:

- (a) I and IV**
- (b) Only IV**
- (c) I and II**
- (d) I, II and III**



3

Q. Calculate the total effective resistance of the circuit.



Handwritten solution for the total effective resistance:

First, calculate the equivalent resistance of the parallel part:

$$\frac{1}{R_{eq}} = \frac{1}{16} + \frac{1}{4}$$
$$= \frac{4+1}{16} = \frac{5}{16}$$

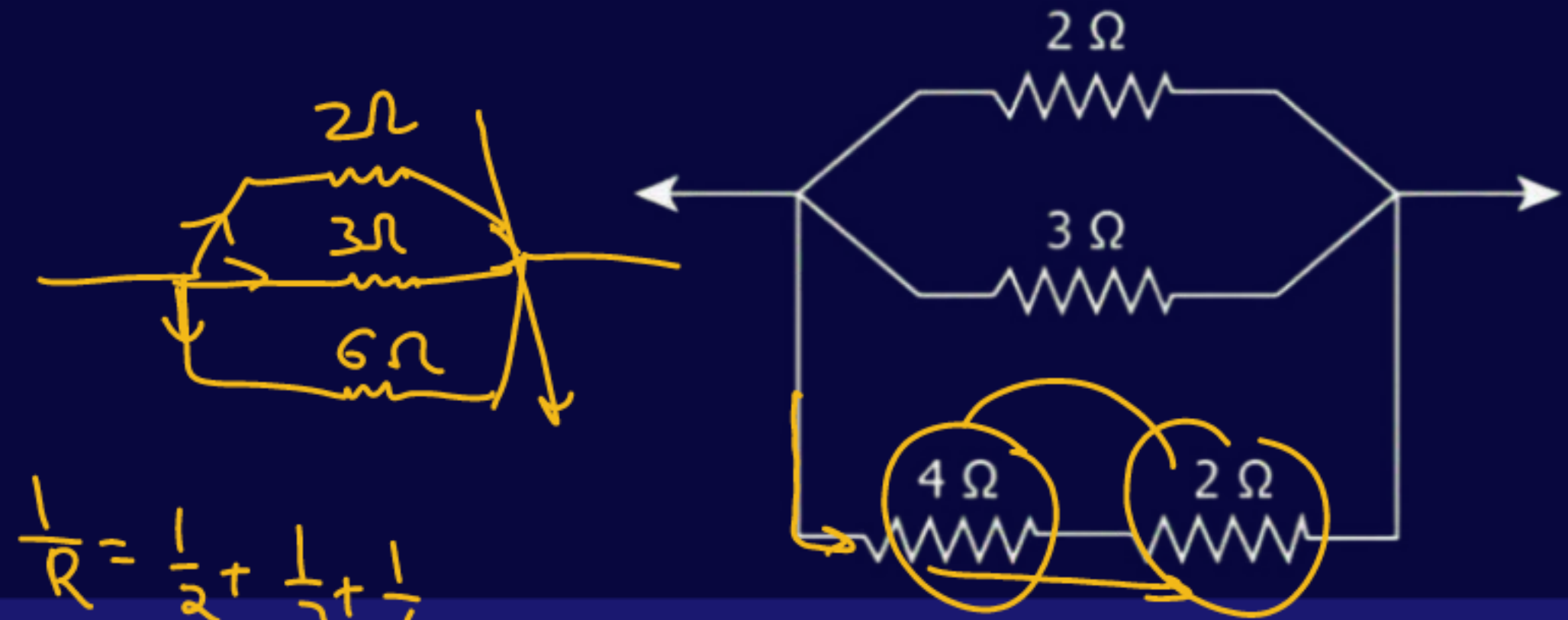
Then, add the series 2 Ω resistor:

$$2\Omega + \frac{16}{5} = \frac{26}{5}\Omega$$

The final answer is $\frac{26}{5}\Omega$.

Q. The image shows a combination of 4 resistors. What is the net resistance between the two points in the circuit?

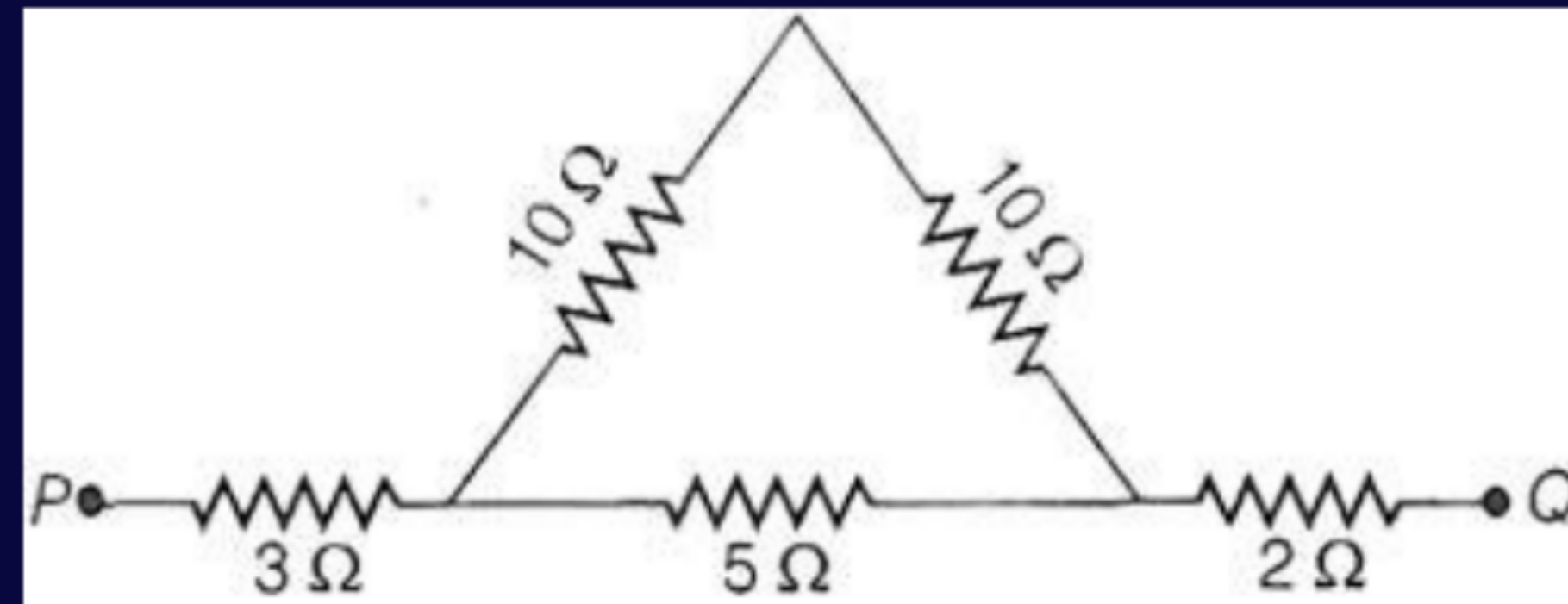
- (a) $0.5\ \Omega$
- ~~(b) $1.0\ \Omega$~~
- (c) $1.5\ \Omega$
- (d) $2.0\ \Omega$



$$\frac{1}{R} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$$
$$= \frac{6}{6} = 1$$

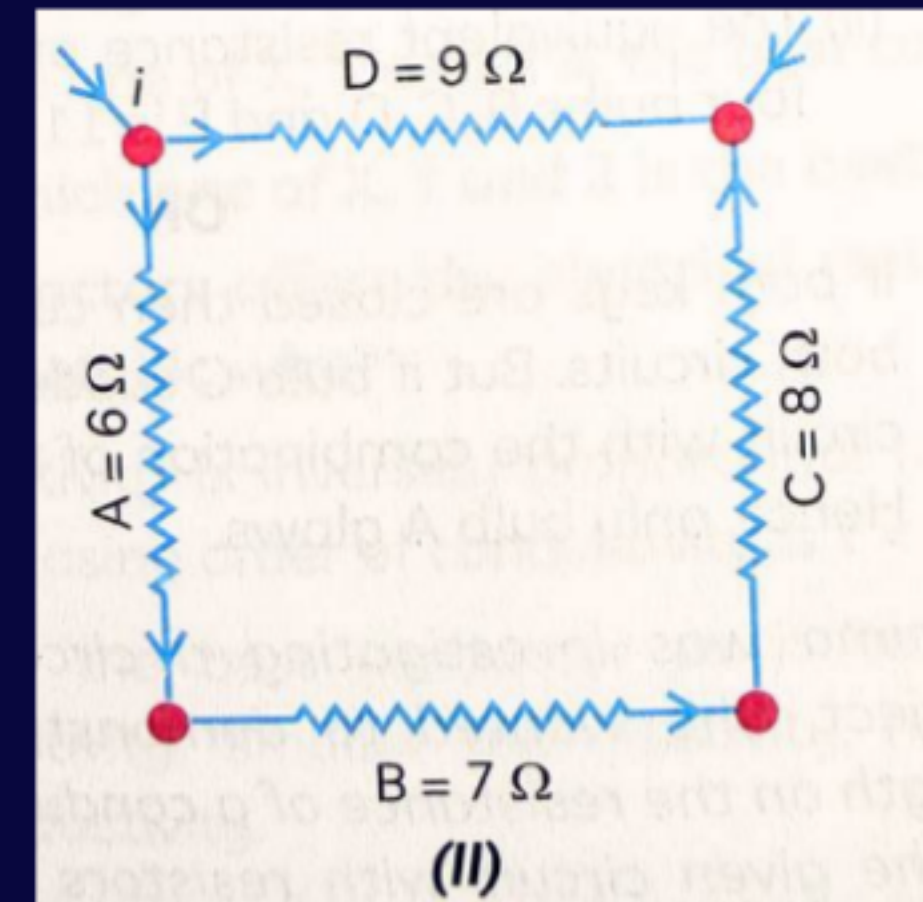
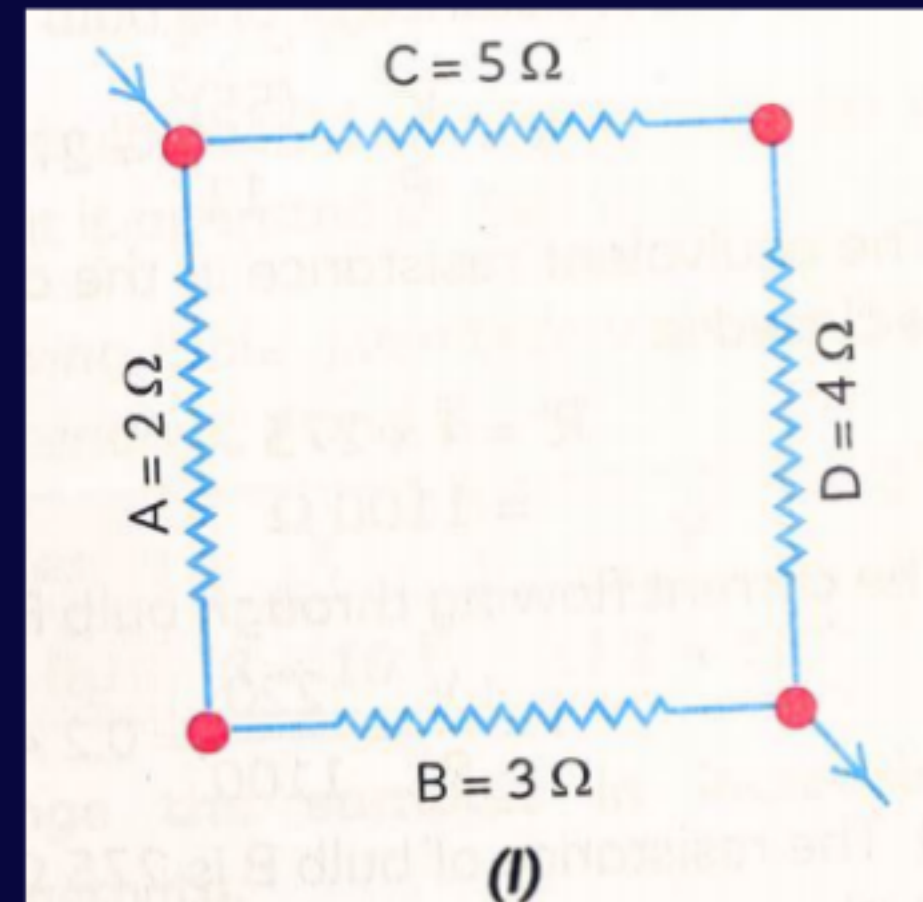
~~Q.~~ Calculate the equivalent resistance

HW

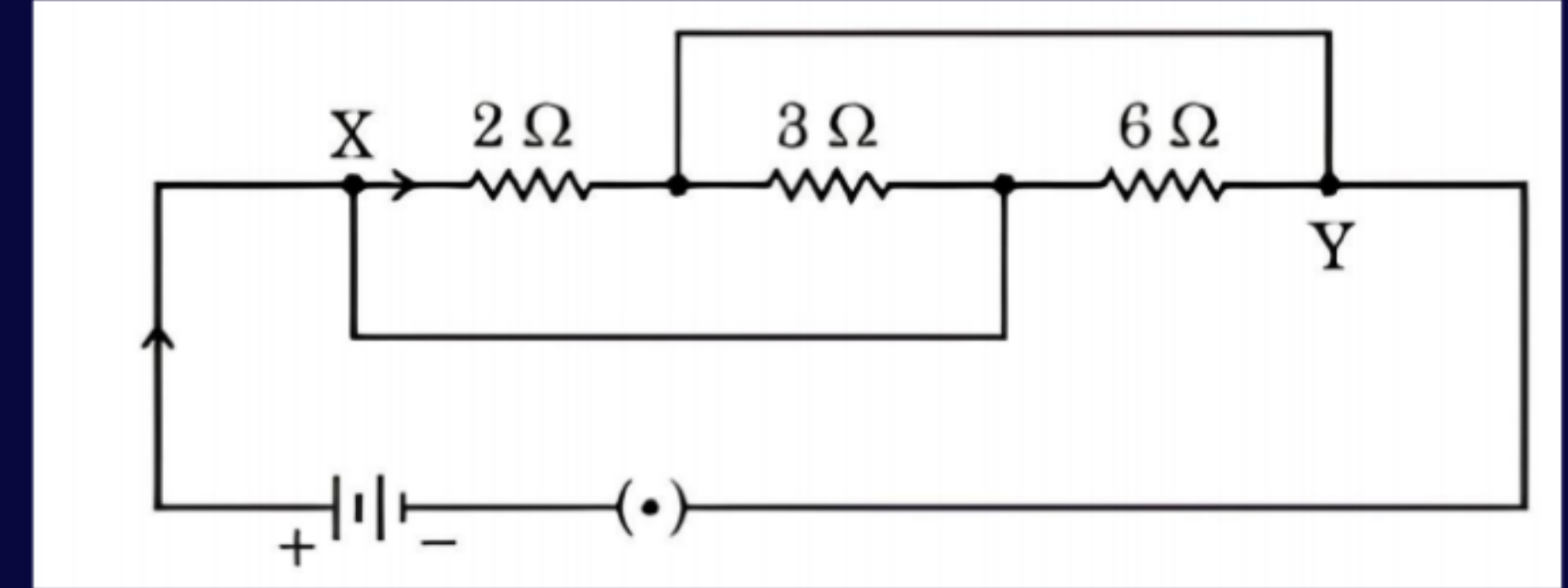


Q. Calculate the total resistance in diagrams (I) and (II).

HW



Q. In the given circuit, the total resistance between X and Y is to be calculated. The circuit contains resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ connected as shown in the figure. Find the equivalent resistance between X and Y.



HW

NCERT

HW

Q. A piece of wire of resistance R is cut into five equal parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R' , then the ratio R/R' is –

- (a) $1/25$
- (b) $1/5$
- (c) 5
- (d) 25

NCERT

Q. How can three resistors of resistances $2\ \Omega$, $3\ \Omega$, and $6\ \Omega$ be connected to give a total resistance of (a) $4\ \Omega$, (b) $1\ \Omega$?

HW

NCERT

Q. What is (a) the highest, (b) the lowest total resistance that can be secured by combinations of four coils of resistance $4\ \Omega$, $8\ \Omega$, $12\ \Omega$, $24\ \Omega$?

HW

Resistors when in series combination



The sum of all resistors is what is my total resistance

Resistors when in parallel combination



Could you please take the reciprocal of all the resistors, Please!